|INCH-POUND|

MIL-H-22547C(YD)
07 July 1989
SUPERSEDING
MIL-H-22547B(YD)
16 July 1976

MILITARY SPECIFICATION

HEAT PUMPS, HEATING AND COOLING (UNITARY) (8,400 TO 300,000 BTU)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air-to-air and water-to-air heat pumps designed to perform the functions of both heating and cooling with controlled temperature, circulating, filtering, cleaning, and dehumidifying of an air stream. The heating functions may be supplemented by electric resistance heaters.

1.2 Classification. The heat pump will be of the following types, classes, and sizes as specified (see 6.2).

Type I - Heat pump, factory assembled refrigerant circuit or circuits (packaged unit).
Type II - Heat pump (split-system) employing remote outdoor section from indoor section.

Class A - Air-to-air.
Class B - Water-to-air.

Size 1 - 8,400 - 65,000 British thermal units per hour (Btu/hr).
Size 2 - 65,001 - 300,000 BTU/hr.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 4120

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1.2.1 Part or identifying numbers (PIN). The specification number, type code, class code, and size code numbers are combined to form the specification PIN for the heat pumps covered by this specification (see 6.5). The PIN for the heat pump units are established as follows:

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                                  M22547 - X - X - X
                                     |     |   |   |
Specification number-----------------     |   |   |
Type code number (see 6.5.2)---------------   |   |
Class code number (see 6.5.3)-----------------   |
Size code number (see 6.5.4)-----------------------
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronics and Associated Equipment).
MIL-R-3593 - Refrigeration and Cooling Equipment (Excluding Household Refrigerators), Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government document. The following other Government document form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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NAVFAC MANUAL

P-89 - Engineering Weather Data.

(Unless otherwise indicated, copies of federal and military specifications, standards and manuals are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Non-Government publications. The following documents form a part of document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (6.2).

AIR CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

ARI 210 - Standard for Unitary Air Conditioning Equipment.
ARI 240 - Standard for Air Source Unitary Heat Pump Equipment.
ARI 270 - Standard for Sound Rating of Outdoor Unitary Equipment.
ARI 320 - Standard for Watersource Heat Pumps.
ARI 325 - Standard for Groundwater Source Heat Pumps.
ARI 340 - Standard for Commercial and Industrial Unitary Heat Pump Equipment.
ARI 380 - Standard for Packaged Terminal Heat Pumps.
ARI 710 - Standard for Liquid-Line Driers.

Directory of Certified Unitary Air Conditioners, Unitary Heat Pumps, Sound-Rated Outdoor Unitary Equipment, Central System Humidifiers.

(Application for copies should be addressed to the Air Conditioning and Refrigeration Institute, 1501 Wilson Blvd., Suite 600, Arlington, VA 22209.)

AIR MOVING AND CONDITIONING ASSOCIATION, INC. (AMCA)

AMCA 210 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 30 W. University Dr., Arlington Hts., IL 60004.)

AMERICAN NATIONAL STANDARDS INSTITUTE INC., (ANSI)

ANSI B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
ANSI B1.20.1 - Pipe Threads, General.
ANSI B1.20.3 - Dryseal Pipe Threads.
ANSI B18.6.4 - Thread Forming and Thread Cutting Tapping Screws and Metallic Drive Screws.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117 - Standard Method of Salt Spray (Fog) Testing.
ASTM B 280 - Seamless Copper Tube for Air Conditioning and Refrigeration Field Service.
ASTM C 534 - Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form.
ASTM D 1654 - Standard Method of Evaluation of Painted or Coated Specimen Subjected to Corrosive Environments.
ASTM D 2247 - Standard Method for Testing Water Resistance of Coatings in 100 Percent Relative Humidity.
ASTM F 872 - Standard Specification for Filter Units, Air Conditioning: Viscous-Impingement Type, Cleanable.
ASTM F 1040 - Standard Specification for Filter Units, Air Conditioning: Viscous-Impingement and Dry Types, Replaceable.
ASTM G 33 - Standard Recommended Practice for Recording Data from Atmospheric Corrosion Tests of Metallic-Coated Steel Specimen.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS, INC., (ASHRAE)

ASHRAE 15 - Safety Code for Mechanical Refrigeration.
ASHRAE 34 - Number Designation of Refrigeration.
ASHRAE 37 - Methods of Testing for Rating Unitary Air Conditioning and Heat Pump Equipment.
Handbook of Fundamentals.

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., 1721 Tullie Circle N.E., Atlanta, GA 30329.)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA M12 - Simplified Procedures for Water Examination.

(Application for copies should be addressed to the Publications Sales Department, American Water Works Association, 6666 W. Quincy Ave., Denver, CO. 80235.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS, INC., (IEEE)

IEEE 112 - Test Procedure for Polyphase Induction Motors and Generators.
IEEE 114 - Test Procedure for Single-Phase Induction Motors.

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(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA DC series - Standards for Residential Controls.
NEMA MG series - Standards for Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Suite 300, Washington, DC 20037.)

NATIONAL ENVIRONMENTAL SYSTEMS CONTRACTORS ASSOCIATION (NESCA)

NESCA Manual J - Load Calculation.

(Application for copies should be addressed to the National Environmental Systems Contractors Association, 1501 Wilson Blvd., Arlington, VA 22209.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 1901 Pennsylvania Ave. NW, Washington, DC 20006.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 353 - Limit Controls.
UL 559 - Heat Pumps.
UL 873 - Temperature-Indicating and Regulating Equipment.
UL 984 - Hermetic Refrigerant Motor-Compressors.
UL 1096 - Electric Central Air Heating Equipment.

Electrical Appliance and Utilization Equipment List.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. Throughout this specification, the term "heat pump" shall be used interchangeably with the term "unitary heat pump" as defined in ARI 240.

3.1.1 Type I. The type I heat pump shall be a complete self-contained electrically operated unit, employing factory assembled refrigeration circuit or circuits. The unit shall utilize the heat content available from a water source or air for the heating cycle, and shall utilize water or outside air for cooling the condenser assembly for the cooling cycle. The unit shall include means for air circulating, cooling, and air temperature controls. Air cleaning means shall be provided and shall be an integral or nonintegral part of the unit as required. The heat pump shall consist of a factory assembled sheet metal enclosure complete with compressor motor assembly, compressor motor starter (when required), compressor motor overload protective devices, and crankcase heaters, (except class B units installed indoors). The class A unit shall employ outdoor coil and outdoor coil fan motor assembly (except water-to-air for class B), an indoor coil and fan motor assembly, an indoor coil fan motor controller or switch and fan overload protective devices (where required). The unit shall be equipped with water-to-refrigerant heat exchanger (for water-to-air for class B), refrigerant and fan overload protective device control devices, changeover valves and check valves (when required), refrigerant and oil charge, a pressure relief means. Supplemental electric heaters shall be supplied when required. Temperature limiting devices shall be furnished on heat pumps with supplemental electric heaters and pressure limiting and accessories required for a complete operating unit shall be furnished. The unit shall be factory charged with refrigerant and oil and ready for operation after being set in place and connected to duct system, water piping (water-to-air), drain, and electrical current. A room indoor thermostat shall be supplied.

3.1.2 Type II. The type II heat pump shall consist of two complete package units. One of the units shall be an outdoor section (remote outdoor coil or water-to-refrigerant heat exchanger in one assembly) and the other an indoor section (indoor coil and fan-motor unit in a second assembly) with the compressor placed in outdoor section. The units shall consist of same component parts as type I unit so placed and arranged that the units may be used together when specified (see 6.2), with interconnecting prefabricated factory charged tubing. An outdoor thermostat shall be provided integral with, or remote from, outdoor unit.

3.1.3 Certain requirements. In this specification, particularly current overload protection, crankcase heating, time delay mechanism, and liquid flood-back protection are included for system reliability needs. If a manufacturer has alternative means to accomplish these functions, the acceptance shall be dependent on proof submitted by the contractor, and be

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acceptable to the authorized Government representative. This proof shall include written certification as part of the first article from the heat pump manufacturer that the alternative means satisfy the functional requirements.

3.1.4 Energy conservation. The contractor shall submit the requirements for energy conservation application data as part of the first article, which shall be developed by the manufacturer of the heat pump equipment in accordance with the criteria and procedures specified in 3.24.6 herein.

3.1.5 Thermostats. All room thermostats shall be of the energy conservation type hereinafter specified (see 3.23.3.3), no other will be permitted, and unless otherwise approved by the contracting officer shall be low voltage electric. All outdoor thermostats shall be of the type here in before specified and shall be mounted in the outdoor unit or remotely mounted as required by the heat pump manufacturer. All thermostats shall conform to UL 873, as applicable.

3.2 Standards compliance.

3.2.1 Certification. All heat pump equipment shall conform to the standards of UL 559, ASHRAE 15, ARI 240, 270, 320, 325, 340, and 380, as applicable, and shall be listed in UL Electrical Appliance and Utilization Equipment List. Materials and equipment shall be new and bear the ARI label, the UL label or ETL Equivalent, or be listed in UL Building Materials Directory or Electrical Appliance and Utilization Equipment List, wherever standards have been established by those agencies. The equipment to be installed by the contractor will equal or exceed the heating and cooling capacity requirements when used together at the design conditions indicated on the drawings. In addition to the foregoing requirements, the submitted heat pump models must comply with the quality assurance provisions of section 4 and tests of 4.4.

3.3 Standard products. Materials and equipment furnished under this specification shall be standard cataloged products of manufacturers regularly engaged in production of such materials or equipment and shall be the standard cataloged product of a manufacturer who has produced that model at least one year prior to the date of bid opening. Items not meeting this requirement, but which otherwise meet technical specifications and the merits of which can be established through the quality assurance provisions and tests required in section 4 will be acceptable. Heat pumps with only minor changes resulting in a model number change, or standard catalog product heat pumps with only changes specifically required to meet this specification may be accepted by the authorized Government representative. The burden of proof that the changes do not adversely affect the performance, reliability, and safety compared to the model for which one year of service has been demonstrated shall rest with the heat pump manufacturer. All heat pumps of a given type, class, and size shall be of the same manufacture and model; however, the component parts of the system need not be the products of the same manufacturer. Each major component of equipment shall have the manufacturer's name and catalog number on a nameplate securely affixed in a conspicuous place; nameplate of a distributing agent only will not be acceptable. Belts,

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pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts located so that a person can come in close proximity thereto, shall be fully enclosed or properly guarded. Unit components shall be readily accessible for maintenance and service.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1, 6.2, and 6.4).

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein (see 6.2), all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5.1 Component materials. Materials used in fabricating component parts of the heat pump and its auxiliaries shall be suitable for use with specified refrigerant, water, and air. No material shall be used that will deteriorate due to chemical action of the fluid specified.

3.5.2 Corrosion protection. All materials of which the components in the outdoor unit are fabricated and the method of fabrication shall be such that when the heat pump is subjected to the reliability tests (see 4.4.4), it shall pass the tests and retain its physical and structural integrity. The use of dissimilar metals in intimate contact with each other when these joints are exposed to moisture in normal operation in order to minimize or eliminate the galvanic effect on coils and fins of the outdoor coil shall be corrosion resistant or treated to resist corrosion. All materials in the outdoor sections shall be of such type or construction that the unit can satisfactorily pass the applicable salt spray and humidity test specified (see 4.4.4.3.1 and 4.4.4.3.2). Units fabricated from ferrous metals, shall be treated for prevention of rust by a coating or paint system that will withstand 125 hours in a salt-spray fog test except the equipment located outdoors shall be tested for 500 hours. The salt-spray fog test shall be in accordance with ASTM B 117 using a 20-percent sodium-chloride solution. Immediately after completion of the test, the coating shall show no signs of rust creepage beyond 1/8 inch on either side of the scratch mark. The film thickness of the factory coating or paint system applied on the equipment shall be not less than the film thickness used on the test specimen.

3.6 Interchangeability. All heat pumps of the same classification fur nished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.7 Lifting attachments. When specified (see 6.2), suitable lifting attachment plate(s) shall be provided to enable the equipment to be lifted in its normal position. Attachment eye(s) shall be not less than 3 inches in diameter, and plate(s) shall withstand any handling conditions encountered, such as rapid lowering and braking of the load. When practicable, only one attachment shall be used. Where more than one is required: (1) each attachment, when practicable, shall be of sufficient capacity to carry the total weight; and (2) a copper- or zinc-base alloy, or aluminum plate, securely fastened to the unit, shall indicate diagrammatically the lifting capacity of each attachment.

3.8 Lubrication. All bearings (except sealed-for-life type), gears, and sliding parts shall be provided with means for lubrication.

3.9 Design. Design of the equipment shall be in accordance with the best engineering practice. The equipment design shall permit accessibility for maintenance and service. The design shall be such as to prevent conditions which may be hazardous to personnel or detrimental to equipment and shall conform to the safety requirements of the UL 559. All screw threads shall conform to ANSI B1.1, B1.20.1, B1.20.3, and B18.6.4 as applicable.

3.10 Construction. The equipment shall be constructed and certified in accordance with 3.2.1. Unless space conditions cause limitations to be specified (see 6.2), all overall dimensions (length, width, and height), shall be as determined by the manufacturer. All replaceable parts shall be constructed to definite standards, tolerances, and clearances in order that any such part of a particular type or model may be replaced without requiring modification.

3.10.1 Couplings. Couplings shall be brazed, sweat fitted, or quick-connect type as specified (see 6.2). Couplings shall be factory fabricated for refrigeration use, having matching attaching ends, and shall be factory dehydrated and furnished with a balanced charge of refrigerant recommended by the manufacturer of the heat pump equipment. Quick-connectors shall completely seal the system against leakage of refrigerant and against all possible entrance of moisture and air into the system, when coupled to equipment. All type II units shall have service valves which allow the refrigerant charge to be isolated in the outdoor unit.

3.11 Fan. Fans utilized in each unit shall be of the centrifugal type or propeller type, quiet in operation and adequately secured to shafts. Shafts shall be supported by at least two self-aligning bearings. Bearings shall be self-oiling with adequate oil reservoirs or permanent life lubricated ball or sleeve bearings. Oil or grease fittings, when used, shall be readily accessible upon removal of the service panel. Fan may be direct- or belt-driven. Where fan is belt-driven, motor shall be provided with adjustable means for V-belt drive. All fans shall be dynamically and statically balanced. Supply air fans shall be centrifugal type capable of supplying the cubic feet per minute (cfm) and static pressure specified (see 6.2.). Air supply fan and motor shall be capable of supplying the specified

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cfm when operating during heating and cooling. Fan ratings shall conform to AMCA 210. V-belt drives shall conform to RMA and have a service factor not less than 1.2.

3.12 Enclosures. The enclosures for the heat pumps shall be in accordance with UL 559. Enclosures intended for outdoor use shall have a galvanized steel thickness not less than 0.034-inch (galvanized standard gauge No. 20), and shall be so constructed as to prevent the wetting of live parts. The outdoor enclosures shall protect the system against shock hazards during weather exposure. The indoor section enclosure shall be thermally and acoustically insulated. When specified (see 6.2), mechanical features shall be provided which shall permit floor, ceiling, or other mounting provisions as required by the procuring activity. Provisions shall be made for easy access to all parts requiring service.

3.12.1 Insulation. To prevent excessive heat transfer and condensate leakage through the cabinet, and reduce noise, the heat pump shall be adequately insulated as required in ARI 240, 320, 325, 340, and 380 as applicable.

3.12.2 Grilles and duct connections. When specified (see 6.2), outlet grille shall be furnished. Grilles and duct connections shall be finished to harmonize with the cabinet. Outlet grilles shall be constructed to permit adjustable directional air flow in both horizontal and vertical planes.

3.12.3 Duct adapters. Duct openings to the outside shall be provided with screens not more than 1/2 inch mesh opening size on the weather size. Provisions shall be made for draining or excluding rain water to prevent its entrance into the unit or room.

3.13 Drain pans. Drain pans shall be provided with corrosion-resistant pipe connections. Metal pans shall be coated with thermal insulation to prevent casing condensation. Plastic reinforced fiberglass pans may be provided if thickness is at least 1/10 inch.

3.14 Air filters. Air filters shall be disposable or permanent type as specified (see 6.2). Filters shall be integral with unit or external with filter rack. Filters shall be accessible for removal or inspection. Filters shall be not less than 1-inch thick, and shall be type I or type II, conforming to ASTM F 1040. Permanent type filters shall be washable, viscous impingement type conforming to type I of ASTM F 872.

3.15 Hermetic motor-compressor units. Hermetic motor-compressor units shall conform to the standards of UL 984 and UL 559 with regard to construction performance, and testing. The requirements for motor-compressor protection systems of these standards shall also apply in addition to the following.

3.15.1 Thermal protection. An internal motor-winding temperature-sensing protection device of the automatic reset type for the compressor motor shall be provided to prevent motor operation under excessive motor winding temperatures. For size 1 heat pump, the manufacturer's recommended maximum motor winding temperature shall not be exceeded when the heat pump is

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operating under any of the tests prescribed in ARI 240, 320, 325, 340, or 380 as applicable, and in no case shall the motor-compressor enclosure temperature under any of these test conditions exceed 300 degrees Fahrenheit (^{o}F) (149 degrees Celsius (^{o}C)) when measured by the resistance method set forth in IEEE 112 or IEEE 114, as applicable. For size 1 heat pumps, as part of the first article tests, a certificate of compliance shall be furnished stating the manufacturer's recommended maximum motor winding temperature.

3.15.2 Current overload protection. Motor running current overload protection of the manual or automatic reset type conforming to UL 984 shall be provided and installed in accordance with NFPA 70.

3.15.3 Crankcase heating. Means shall be provided for heating the compressor crankcase during "OFF" cycles to minimize liquid foaming, and prevent consequent liquid sludging under compressor starting conditions caused by retention of refrigerant by the oil in the compressor. Crankcase heaters either external or internal (except for motor windings used as heaters) shall be readily replaceable from the exterior without breaking the refrigerant seal.

3.15.4 Time delay mechanism. When a compressor or operation shutdown occurs for any reason, inadvertent or otherwise, the motor protection system shall contain a device or means that delays operation of the compressor motor for a period no less than 3 minutes, but not greater than 6 minutes after compressor operation is called for. Unless otherwise specified (see 6.2), the time delay mechanism shall be factory installed.

3.15.5 Liquid flood-back. A means shall be provided for protecting the compressor from liquid flood-back under all design conditions of operation, including defrost. Properly designed and sized internal or external accumulators that allow a free flow of oil back to the compressor and in conjunction with defrost mechanism and crankcase heating will satisfy this requirement.

3.16 Refrigerant. The refrigerant shall be group I classification as described in ASHRAE 34, unless specified (see 6.2), the refrigerant shall be R-22.

3.17 Factory charge. Type I units and type II, size 1 outdoor sections shall be completely factory charged with the amount of refrigerant and lubricating oil recommended by the manufacturer. Type II, size 2 outdoor sections shall be shipped by the manufacturer with a holding charge of refrigerant.

3.18 Refrigerant piping. Couplings shall be brazed, sweat fitted, or quick-connect type as specified (see 6.2). When specified (see 6.2), interconnecting, prefabricated factory charged tubing shall be provided with length in feet. Tubing shall be extra soft, deoxidized, bright annealed copper conforming to ASTM B 280 factory dehydrated and furnished with a balanced charge of refrigerant recommended by manufacturer of the heat pump, and factory insulated suction line with 3/8-inch minimum thickness of

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closed cell, foamed plastic conforming to ASTM C 534 with a permeance rating not to greater than 1.0 perm. Couplings shall be included for suction and liquid line connections of the indoor and outdoor sections.

3.19 Gauge ports and service valves. Gauge ports or charge valves shall be provided for routine and periodic pressure testing of the system and field charging or removing of refrigerant.

3.20 Moisture in refrigerant. The system shall contain a filter drier in the liquid line to both the indoor and outdoor coils, conforming to ARI 710. Filter driers shall be so installed as to be easily removed for replacement. Refrigerant piping shall be circuited such that backwashing of the filter drier by reversal of refrigerant flow does not occur. The moisture content in the refrigerant shall not exceed 60 parts per million (ppm) at time of heat pump installation and 30 ppm when operating with the filter-drier installed and at the conclusion of the reliability tests (see 4.4.4).

3.21 Strainer. If the heat pump system uses capillary tubes as a refrigerant expansion device, strainer(s) or other suitable means shall be installed in the line ahead of each capillary tube assembly capable of stopping passage of particulate matter of minor dimension greater than 20 percent of the bore size of the smallest tube.

3.22 Water-to-air heat pumps. The characteristic requirements for water-to-air heat pumps are:

a. The water temperature leaving the condenser during cooling operation shall be not greater than 100°F (37.8°C).
b. The water temperature leaving the chiller during heating operation shall not be less than 45°F (7.2°C).
c. A low pressure cutout device with a manually reset lockout relay shall be installed near the inlet to the compressor or a freezestat in the water circuit shall be installed near the water outlet of the water to refrigerant heat exchanger, shall be provided to prevent freezing during the heating operation.
d. The water-to-refrigerant heat exchanger, internal piping, and connections, unless otherwise specified (see 6.2), shall be composed of cupro-nickel alloy.
e. Unless otherwise specified (see 6.2) construction and/or piping for water-to-refrigerant heat exchangers shall include provision for isolation and access valving for chemical cleaning or physical access for mechanical cleaning of the water side of heat exchange surfaces. The quantity in gallons per minute (gpm) and temperature for both winter and summer seasons and quality of water available to the heat pump on a continuous basis is as specified (see 6.1.1c and 6.2).

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3.23 Electrical components.

3.23.1 Motor and motor overload protection. Motors and motor overload protection shall be suitable for the electrical characteristics specified (see 6.2). Motors shall be designed for the voltage specified herein. Motor and motor overload protectors shall conform to UL 559 and NEMA MG1. Motor overload protectors for hermetic motor-compressor units shall conform to the requirements of UL 984 and as specified in 3.15. Motor starters shall conform to the applicable requirements of NEMA DC series and NFPA 70.

3.23.2 Electrical heaters.

3.23.2.1 Supplemental electric heating. When supplemental heating is required to satisfy the heating requirements (see 6.2), electrical resistance heaters shall be provided integral with unit located downstream of indoor coil. Unless otherwise specified (see 6.2), capacity in kilowatts (kW), number of circuits, and stages shall be as recommended by the manufacturer to satisfy the heating requirement of 3.24.1 and as specified in 3.23.3.4. When specified (see 6.2), an electrical duct heater shall be provided for remote installation in duct work.

3.23.2.2 Electrical duct heaters. When specified in 3.23.2.1, duct heaters shall be of packaged design arranged for inserting in and connecting to air duct so heating elements are directly in the air stream. The heaters shall have low air pressure drop at nominal air flow requirements or at air flow specified (see 6.2). Electric duct heaters shall be of the open coil or finned tubular type, and shall have heater wires which are adequately protected against corrosion, moisture, or sagging. Duct heaters shall be of rugged construction which protects the resistor wire for life from electrical or mechanical shock, vibration or breakage in shipment, installation, and use. Dummy elements of expanded metal shall be provided where necessary to uniformly distribute air flow across heater face. Electric heaters shall conform to the requirements of UL 1096 and NFPA 70. Number of electrical circuits shall be as specified in 3.23.3.4. Heater wattage rating shall be within 5 percent of the specified rating. Each duct heater shall have two high-limit controls. The primary automatic reset thermal cutout high-limit control shall be wired to the controlling device if possible, and the secondary manual reset thermal cutout high-limit device will be wired to de-energize the main power. Both high limits will be used to protect the elements from excess temperature in case of airflow stoppage.

3.23.3 Controls. Manual and automatic controls, factory wired and accessible for service and inspection shall be provided. Manual controls shall permit operation of the supply fan only; supply fan and heating only; and supply fan and cooling only.

3.23.3.1 Defrost control. Air source heat pumps, designed to operate with a defrost cycle, shall include positive actuation and termination means for defrosting the coils and fins of the outdoor unit automatically with suitable drain provisions to remove condensate and meltage without freeze-up or other malfunctions throughout the entire range of application temperatures.

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3.23.3.2 Protective mechanisms, controls, and auxiliary equipment. The heat pump system shall contain the necessary components, protective mechanisms, controls, and auxiliary equipment to meet the requirements and tests set forth in this specification. No safety device or protective control shall be used as an operating device. All limit controls shall conform to UL 353. Pressure relief safety devices shall be in accordance with ASHRAE 15.

3.23.3.3 Indoor temperature control. Room thermostat shall be of the heating-cooling type with separate temperature sensing elements for each system. Electrical rating shall be 1.5 amperes at 30 volts maximum alternating current (ac). The thermostat shall be designed for a mechanical differential not to exceed 1.5°F (.8°C), shall be without thermometer, and shall be provided with external temperature setting devices with permanent factory installed means to prevent heating temperature setting above 75°F (23.9°C) with an adjustable range of at least 15°F (8.3°C) for heating and cooling circuits, and shall be electrically isolated from each other. The housing shall be corrosion-resistant metal, painted finish, or molded plastic. The heating circuit shall have adjustable or fixed heater for heat anticipation. The cooling circuit shall have fixed cooling anticipation heater. When specified (see 6.2), all heat pumps shall be provided with manual means for temperature setback. The thermostat shall contain a subbase with selection switches for OFF-HEAT-COOL and FAN-AUTO-ON. The reversing valve shall be controlled by manual HEAT-OFF-COOL system switch. The thermostat shall be designed to permit interlock with the outdoor thermostats and operation of supplemental electric heating by remotely located emergency switch. Limiting means shall be designed to prevent field alterations of the setting limit. For space temperatures between 70°F (21.1°C) and 78°F (25.6°C), the automatic control system shall be capable of being set so that neither heating nor cooling is provided. When the heat pump is operating, except during defrost operation, the temperature control shall regulate the indoor temperature at the thermostat to not less than 2°F (1.1°C) of the control setting when only compression heating is being provided, and to not less than 3°F (1.7°C) of the control setting when compression and supplemental heating are required. Control of supplemental heating shall be in accordance with 3.23.3.4. Thermostats shall have a single pole throw switch hermetically sealed and actuated by a bimetallic or bellows type element. Each element shall operate switches to provide single or multistate control for heating and cooling equipment.

3.23.3.4 Supplemental heat control. When supplementary heat is needed to satisfy the heating requirements in compliance with 3.24.1, electric resistance heaters shall be provided. The electric heaters shall be electrically wired in accordance with UL 559, and controlled in one or more steps so that they shall provide sufficient additional heating capacity for building heat demand loads in excess of the compression and indoor fan heating capacity. No single step of supplemental heating shall produce a temperature rise in the discharge air greater than 40°F (4.4°C). The heaters shall be controlled by means of outdoor and indoor thermostats interlocked one with the other. The outdoor thermostat(s) shall be provided by the heat pump manufacturer whose operation shall override that of the indoor thermostat, shall prevent the supplementary heaters from operating at outdoor temperatures greater than 40°F (4.4°C) or as specified (see 6.2), except as required to

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offset cooling effect during defrost operation. For the heating operation, the indoor thermostat shall operate in at least two steps. The first step shall initiate heat pump operation for heating by compression only. The second step (and additional step(s), if provided) shall control heating provided by the supplemental heaters. The cut-off temperature for the first set of supplemental heaters shall be no less than 0.5°F (.3°C) and no greater than 1.5°F (.8°C) below the cut-off temperature for the compression heating. The cut-on temperature of the first set of supplemental heaters shall be no less than 0.5°F (.3°C) and greater than 1.5°F (.8°C) below the cut-on temperature for the compression heating. The total operating temperature differential between cut-on and cut-off of the compression heating control shall be not greater than 2°F (1.1°C). Additional steps of supplemental heating shall be sequenced on as outside air temperature continues to drop. The outside air thermostat shall be set so that compression heating plus supplementary heating equals the required heating load, down to the design outside temperature, without bringing on unnecessary supplementary heating steps. All quoted thermostat temperature differentials are as measured at the switch.

3.24 Performance.

3.24.1 Heat pump cooling and heating capacity. The heat pump heating and cooling (sensible and latent) capacities shall be not less than that specified (see 6.2) for the heating and cooling indoor and outdoor conditions, when rated in accordance with ARI 240, 320, 325, 340, or 380 as applicable. If the sensible cooling capacity of the heat pump at summer design conditions is greater than twice the design sensible cooling load, the heat pump shall be equipped with means to modulate or otherwise limit the heat pump cooling capacity-to-design-cooling-load ratio to no more than 1.3. The heating-by-on compression capacity shall be such that the heating seasonal performance factor (HSPF) or seasonal performance factor (SPF) when required in 3.24.5 is met. Unless otherwise specified (see 6.2), the minimum capacity for the supplemental heaters shall be 80 percent of the heating load. The heat pump manufacturer shall state the rated voltage and needed capacity of supplemental heaters. The heaters shall provide this capacity when tested in accordance with 4.4.2.

3.24.2 Sound rating. All outdoor unitary heat pump equipment shall be sound rated and tested in accordance with ARI 270 as applicable and shall be not greater than a sound rating of 8.4 bels for units below 38,000 Btu/hr (11,100 watts), 8.6 bels for units between 38,000 Btu/hr (11,100 watts) and 65,000 Btu/hr (19,000 watts) and 8.6 bels for units greater than 65,000 Btu/hr (19,000 watts) in capacity as defined in these standards. ARI listing of complying sound ratings for the submitted models and sizes and ARI labels signifying ARI certified sound ratings in accordance with ARI 270 will be acceptable as sufficient evidence of compliance. In lieu of such certification or label, a written certificate as part of the first article from a testing agency or laboratory adequately equipped and competent to perform such services shall be submitted stating that the equipment has been tested and conforms to requirements specified in ARI 270.

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3.24.3 Operational. The heat pump system shall operate satisfactorily under specified normal and maximum operating conditions in accordance with ARI 240, 320, 325, 340, and 380 requirements as applicable.

3.24.4 Reliability. The ability of the heat pump system to withstand conditions that might be encountered in service without damage to any part of the system shall be demonstrated as required in 4.4, when specified (see 6.2).

3.24.5 Energy performance criteria. Heat pump energy consumption and efficiencies shall be in accordance with paragraph 3.24.5.1, 3.24.5.2, or 3.24.5.3, as applicable.

3.24.5.1 Energy performance requirements for all class A (air-air), 3 phase power heat pumps. The minimum Energy Efficient Ratio (EER) for cooling when rated in accordance with ARI 240 or 340 as applicable shall be not less than 8.0.

3.24.5.2 Energy performance requirements for class A (air-air), size 1 (through 65,000 BTUH), single-phase-power heat pumps. The heat pumps shall have the following minimum Seasonal Energy Efficiency Ratios (SEERs) and minimum HSPFs when rated in accordance with ARI 240.

	Minimum SEER	Minimum HSPF	Minimum SEER + HSPF
TYPE I (pkg)	8.8	5.8	15.5
TYPE II (split)	9.0	6.0	16.0 *

* Example: Either 9.0 SEER + 7.0 HSPF = 16.0 or 10.0 SEER + 6.0 HSPF = 16.0 would be acceptable. 10.2 SEER + 5.8 HSPF = 16.0 would not be acceptable.

3.24.5.3 Energy performance criteria for class B (water-air) heat pumps. Water source heat pumps shall have a minimum cooling EER of 9.0 and a minimum heating coefficient of performance (COP) of 3.0 when rated in accordance with ARI 320 or ARI 325. Seasonal performance factors, if specified (see 6.2), shall be minimum SPF's.

3.24.6 Energy conservation data. The SPF shall be used in accordance with the performance requirements for Class A, (air-air), 3 phase power heat pumps.

a. Criteria for energy conservation - SPF. A sample calculation for for SPF is shown on table I. Air source heat pumps shall have SPF for the heating season, including supplement heating and frosting of:

Outdoor design temperature, degrees F	SPF (minimum)
35 to 45	2.25
20 to 34	2.0
17 to 19	1.8

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b. Procedure for determining the SPF: The SPF for the heating season shall take into account all system energy requirements, including defrost operation, all fan motors and supplemental heat and shall be calculated from the relationship.

$$SPF = \frac{X}{Y + Z}$$

Where X = heat delivered to building during heating season, kilowatt hour (kWh)
Y = energy input to all motors during heating season, kWh
Z = supplemental heat during heating season, kWh

Values for X, Y, and Z shall be calculated in tabular form from the following information.

A - Outdoor temperature range, degrees F; outdoor air temperature increments of 5°F below 65°F.
B - Seasonal heating hours for A, hours; found in AFM 88-8, Chapter 6 as specified (see 6.2), or is available from the U.S. Weather Bureau.
C - Average building heat loss for A, Btu/hr (including outside air load, if any).
D - Average heat pump capacity for A, Btu/hr; compressor and indoor fan heating, no supplemental heating.
E - Percent running time for heat pump, percent; if C > D, E = 100, otherwise E = 100 x C/D.
F - Rate of energy input to heat pump for A, kW (based on actual entering air conditions at indoor coil including effects of outside air load if any).
G - Energy input to heat pump, kWh; G = B x E x F/100.
H - Energy input from electric resistance heaters, kWh; if C < D, H = 0; otherwise H = B x (C-D)/3412.
J - Energy input to building from heat pump, kWh; J = B x E x D/341200.

Summing over the outdoor air temperature increments gives

$$X = J + H$$
$$Y = G$$
$$Z = H$$

for substitution in the definition for SPF.

c. Sample calculation: A sample calculation for SPF is illustrated for a location at Forbes AFB, Kansas, the seasonal heating hours of column B are found in AFM 88-8, Chapter 6, and the winter

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design temperature is 3°F. The heating load is assumed to be 90,000 Btu/hr with an indoor temperature of 70°F and a design temperature difference relationship:

$$Q = \frac{90,000}{70-3} \quad (65-T)$$

where T is the average temperature for the temperature range in column A. For illustrative purposes only, columns D and F contain assumed values for the performance of the heat pump at the temperature range listed in column A. In determining the SPF of each type, class, and size of heat pump unit submitted by the contractor, the manufacturer of the heat pump shall calculate the the SPF for the published, certified performance data at the temperature range identified by column A. Performing the indicate calculations given above, an SPF equal to 2.09 is found.

TABLE I. Sample calculation for SPF.

A	B	C* x 3 10	D** x 3 10	E	F**	G	H	J
60/64	675	4.0	75.2	5.3	6.57	235.9		791.3
55/59	585	10.7	71.6	14.9	6.30	550.8		1834.6
50/54	568	17.5	67.5	25.9	6.02	886.5		2913.2
45/49	542	24.2	63.4	38.2	5.75	1189.6		3844.2
40/44	572	30.9	59.0	52.4	5.43	1626.7		5180.2
35/39	600	37.6	49.5	76.0	5.15	2347.2		6612.0
30/34	612	44.3	45.2	98.0	4.86	2915.1		7946.0
25/29	417	51.0	41.5	100.0	4.60	1918.2	1161.0	5071.9
20/24	326	57.8	37.2	100.0	4.40	1434.4	1968.2	3554.3
15/19	222	64.5	34.0	100.0	4.23	939.1	1984.5	2212.2
10/14	155	71.2	30.5	100.0	4.10	635.5	1848.9	1385.6
5/9	96	77.9	27.6	100.0	3.90	374.4	1415.2	776.6
0/4	47	84.6	24.0	100.0	3.77	177.2	834.8	330.6
-5/-1	20	91.3	21.0	100.0	3.62	72.4	412.1	123.1
-10/-6	3	98.1	18.0	100.0	3.51	10.5	70.4	15.8
						= 15313.5	9695.1	42591.6

$$SPF = \frac{42591.6 + 9695.1}{15313.5 + 9695.1} = 2.09$$

$$\text{Percent heat by compression} = \frac{42591.6 \times 100}{42591.6 + 9695.1} = 81.4$$

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* Including outside air load, if any
** Based on actual entering conditions at indoor coil, including effects of outside air load, if any.

NOTE: Columns C and D are based on midpoint of temperatures in column A (e.g., 62, 57, 52, 47, 42 --- etc. ^{o}F).

3.24.7 Minimum external duct resistance. Unless otherwise specified (see 6.2), indoor conditioning equipment shall be designed to work against the minimum external resistances established in ARI 240 when delivering the rated capacity and air quantity specified.

3.25 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.26 Treatment and painting. Unless otherwise specified (see 6.2), the equipment shall be treated and painted in accordance with the manufacturer's best standard practice. Outdoor equipment shall conform to 3.5.2.

3.27 Electromagnetic compatibility. When specified (see 6.2), the heat pump shall be designed and equipped for electromagnetic compatibility in accordance with class IIB equipment of MIL-STD-461 (see 4.5).

3.28 Identification marking. Equipment assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 to include the following:

a. Name of the organization sponsoring the completely assembled air conditioning assembly.
b. Type, model or catalog number, serial numbers, and the name of the manufacturer of compressor, controls, and motors sufficient for identification, replacement of parts, or servicing.
c. Date of manufacture, and length of warranty.
d. Starting (locked-rotor) current and running current; if ac, the frequency and phases; if dual voltage, a diagram for changing voltage shall be shown.
e. The refrigerant used, expressed in chemical or other descriptive terms and amount of refrigerant required.
f. Factory test pressure.
g. Location of the ventilating damper, when provided, and of the control switch and thermostat, properly marked or distinctly indicated.
h. Instruction plates, if applicable (see 3.28.1).
i. When specified (see 6.2), the marking shall include the Federal Stock Number.
j. Appropriate UL listing mark.

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3.28.1 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.29 Spare parts and maintenance tools. When specified (see 6.2), spare parts and maintenance tools shall be furnished.

3.30 Workmanship. The quality of workmanship shall be such as to produce heat pumps that are in accordance with the requirements of this specification and are so constructed as to ensure proper functioning of all parts of the unit.

3.30.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.30.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.30.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.30.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.30.5 Machine work. Tolerances for contact and bearing surfaces shall conform to standards prevailing among manufacturers normally producing heat pumps.

3.30.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one heat pump when a first article is required (see 3.4 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4 and 4.6, and the packaging inspection of 4.7. The inspection shall be performed on the samples selected in accordance with 4.2.1.

4.2.2.1 Inspection lot. All units of the same type, style, and capacity, offered to the Government at one time, shall be considered a lot for purposes of inspection. The sample unit shall be one complete heat pump.

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4.2.2.2 Sampling for examination. Sampling and inspection procedures shall be in accordance with MIL-STD-105 (see 6.2). The unit of product shall be one heat pump of each type, model, and size scheduled for delivery. All heat pumps offered for delivery at one time shall be considered a lot for purpose of inspection.

4.2.2.3 Sample for capacity and operational tests. When specified (see 6.2) in the contract, one heat pump of each type, model, and size scheduled for delivery and picked at random shall have been or shall be subjected to all listed capacity tests and operational tests and shall satisfactorily conform to all test requirements. The tests on each heat pump shall be made sequentially with the capacity tests being performed first. The heat pump tested as representative of each type, model and size shall pass all tests described in 4.4.2 and 4.4.3.

4.2.2.4 Sampling for reliability tests. When specified (see 6.2) in the contract, at least one heat pump of each type, class, model and size scheduled for delivery and picked at random shall be subjected to and successfully pass the reliability tests (see 4.4.4), and shall be conducted in accordance with the following plan. If there are no failures in the first sample, the type, class, model, and size will be considered satisfactory with respect to the reliability tests and the unit tested can be included in the lot for delivery. If there are one or more failures, a second trial, using three additional systems of the same type, class, model, and size picked at random by a Government inspector shall undergo a test for conformance to all listed reliability requirements. If there is not more than one failure of the same kind, or not more than two failures altogether from that total of four systems tested, the lot will be considered satisfactory with respect to the reliability tests. Any unit failing shall not be included in the lot for delivery. If there are two or more failures of the same kind, or three or more failures altogether from the total of four systems tested, all heat pumps of that type, class, model, and size shall be rejected.

4.3 Examination. Each heat pump shall be examined for defects listed in table II. Each attribute within each classification of multiple defects shall constitute on defect.

4.4 Tests. Each sample heat pump shall be tested and any unit failing to pass the following tests, as applicable, shall be rejected. Certified testing from an independent laboratory approved by the contracting officer will be accepted as evidence that the heat pumps have passed the required tests. Tests shall be conducted as outlined in the referenced documents as herein specified.

4.4.1 Tests for quality conformance. Three types of tests required by this specification shall be performed, or shall have been performed for each type, class, model, and size of heat pump to be supplied in the contract; namely, (1) capacity, (2) operational, and (3) reliability tests. The data required in this paragraph shall be derived from tests described in 4.4, with regard to all inspections and capacity, operational, and reliability tests. The Government reserves the right to make any or all inspections and tests necessary in addition to those required in the contract to assure conformance with specification requirements at any time prior to final acceptance. If such

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TABLE II. Classification of defects.

Classification	Defects	Requirement Paragraph
Critical:	None defined	
Major:		
101	Thermostats not as specified	3.1.4
102	No evidence of conformance to ASHRAE 15 and ARI 240 and UL 559	3.2.1
103	Material not as specified	3.5
104	Interchangeability of component parts, assemblies, accessories, or space parts not as specified	3.6
105	Lifting attachments not as specified	3.7
106	Means for lubrication not as specified	3.8
107	Design not as specified	3.9
108	Threaded parts not as specified	3.9
109	Construction not as specified	3.10
110	Fan not as specified	3.11
111	Enclosures not as specified	3.12
112	Drain pans not as specified	3.13
113	Air filters not as specified	3.14
114	Hermetic motor-compressor units not as specified	3.15
115	Refrigerant not as specified	3.16
116	Size 1 packaged units not factory charged	3.17
117	Refrigerant piping on split systems of size 1 units not as specified	3.18
118	Gauge parts and service valves not provided	3.19
119	Filter-drier not provided	3.20
120	Strainer not installed correctly when heat pump system uses capillary tubes	3.21
121	Characteristic requirements for water-to-air heat pumps not provided	3.22
122	Electrical components not as specified	3.23
123	Supplemental heaters not as specified	3.23.2
124	Controls not as specified	3.23.3
125	Fungus resistance not as specified	3.25
126	Treatment and painting not as specified	3.26
127	Electromagnetic compatibility not as specified	3.27
128	Workmanship not as specified	3.30
Minor:		
201	Identification marking missing, incomplete, or not legible	3.28
202	Instruction plates not as specified	3.28.1
203	Spare parts and maintenance tools not as specified	3.29

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additional testing indicates failure to conform, the cost of the unit or component tested and the cost of testing will be borne by the contractor. If additional testing indicates conformance, the cost of the unit or component tested and the cost of testing will be borne by the Government.

4.4.2 Capacity tests. This section describes tests of heat pumps for determination of capacity, SEER, HSPF, EER and COP. All of these tests shall be performed without the indoor thermostat in the system or with the thermostat shunted to call for continuous operation of the compression system. Both cooling and heating tests shall be performed as described in the following paragraphs. Testing procedures shall be as described in ASHRAE 37. Unless otherwise specified (see 6.2), test conditions shall be as set forth in ARI 240, 320, 325, 340, and 380 as applicable.

4.4.2.1 Capacity tests - heating. For air-source heat pumps, the contractor shall establish curves of heating capacity and power input (including defrost effects, but not including supplemental heat) versus outdoor air temperature. The outdoor temperature range should be from 62°F (16.7°C) to the winter outdoor design temperature specified in the contract, or to the lowest temperature at which the heat pump will operate to satisfy the SPF calculation required in 3.24.1. Tests shall be performed for the heating capacity and power input at the standard rating condition as described in ARI 240, 320, 325, 340, and 380 as applicable. In addition, tests shall be performed at the application rating conditions as described in ARI 240, 320, 325, 340, and 380 as applicable. For water-source heat pumps, tests shall be performed for the heating capacity and power input versus entering water temperature and minimum water flow (or leaving water temperature) if the entering winter design water temperature specified in the contract is less than 60°F (15.6°C). In addition, tests showing heating capacity and power input shall be performed for the standard rating conditions described in ARI 240, 320, 325, 340, and 380 as applicable.

4.4.2.2 Capacity tests - cooling. For air-source heat pumps, tests showing the cooling capacity and power input shall be performed for the design summer outdoor temperature. Tests shall be performed for the standard rating conditions as described in ARI 240, 320, 325, 340, and 380 as applicable. In addition, tests shall be performed at the application rating conditions as described in ARI 240, 320, 325, 340, and 380 as applicable. For water source heat pumps, tests shall be performed for cooling capacity versus entering water temperature and minimum water flow rate (or leaving water temperature) if the entering summer design water temperature specified in the contract is greater than 75°F (23.9°C). In addition tests shall be performed for the cooling capacity at standard rating conditions as described in ARI 240, 320, 325, 340, and 380 as applicable.

4.4.2.2.1 Air system distribution resistance. For the preceding heating and cooling capacity tests, if in the contract, an air system distribution resistance or air flow different from the values listed in ARI 240, 320, 325, 340 and 380 as applicable, is specified 3.24.7, the performance data required shall be adjusted accordingly.

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4.4.2.3 Performance tests. At standard rating conditions specified in ARI 240, 320, 325, 340, and 380, the SEER, HSPF, EER, and COP, as applicable shall be calculated from the test data, and shall satisfy the performance criteria in 3.24.5.

4.4.2.4 Heating capacity of supplemental heater. The heating capacity of the supplemental heaters shall be determined by the measurement of the electrical power usage of the supplemental heater(s). The electrical power usage shall be measured at specified voltage and at room temperature of 70°F with suitable instrumentation accurate to within 2 percent of the measured value. The test data shall be used to determine compliance with 3.23.2.

4.4.3 Operational tests. The heat pump system shall operate satisfactorily under specified normal and maximum operating conditions. As specified in 3.24.3, heat pumps scheduled for delivery under this specification shall meet all performance requirements of ARI 240, 320, 325, 340 and 380 as applicable. The ARI label shall be accepted as evidence of meeting this requirement. In lieu of such certification or label, testing may be conducted by a testing agency or laboratory adequately equipped and competent to perform such services in accordance with the requirements specified in ARI 240, 320, 325, 340, and 380 as applicable. The tests required in ARI 240 are as follows:

a. Maximum operating condition test.
b. Voltage tolerance test.
c. Low temperature operation test for cooling.
d. Insulation efficiency test.
e. Automatic defrosting test.

4.4.4 Reliability tests. When specified (see 6.2), the heat pump shall be subjected by the manufacturer to a high stress and a life cycle reliability test, and shall conform to the listed requirements, and shall perform without damage to any part of the system shall be demonstrated as specified in 3.24.4. The high stress test program is intended to take the system beyond its normal operating limits to ensure that there are adequate safety devices that will shut the system down in order to protect the essential, more expensive components (i.e., compressor). The life cycle tests are designed to have the unit operate under simulated field conditions near or slightly beyond the worst expected climatic conditions for an extended period of time. In addition to either of these test programs when specified (see 6.2), one or both of the corrosion tests described in 4.4.4.3 must be completed; depending on the climatic conditions of the installation site.

4.4.4.1 High stress reliability tests (table III). Temperature tolerances for the reliability tests shall be as specified in performance requirements of ARI 210/240. Voltage tolerances shall be ±3 volts. All of these tests shall be performed without the indoor thermostat in the system or with the thermostat shunted to call for continuous operation. All manual-reset operating, protective, and limit controls/devices shall be reactivated within 30 minutes following trip-out or similar operation throughout each test. Supplemental heaters shall not be operated in any of the reliability tests. Indoor unit air inlet temperature shall be held in the range not less than 70°F (21.1°C) or not greater than 80°F (26.7°C) for tests. For all tests, the

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portions of the tests that call for unit operation imply continuous compressor operation except as controlled by protective/limit devices. High stress reliability tests (see table III) for water source heat pumps shall be performed with water, not freezing inhibited brine, as the circulating fluid. As applicable, the following requirements must be met for acceptance.

a. No impairing damage shall occur to any component during these tests.
b. The temperature of the oil sump remote from the suction line inlet to the compressor shall at no time during steady-state operation be less than 40°F (22.2°C) warmer than that of the suction line within 6 inches (0.152m) of the compressor shell. During the off cycle, this temperature difference may decrease to 10°F (5.6°C). On start up, this temperature difference must constantly increase toward its steady-state value.
c. The temperature of the discharge line within 6 inches of the compressor shall at no time during operation (for more than 5 minutes) of the compressor motor be less than 20°F (-6.67°C) warmer than the saturation temperature corresponding to the discharge pressure. Thermocouples shall be used for these measurements, shall be physically bonded to the respective surfaces (including at least 1 inch of the leads) and shall be insulated from ambient conditions.
d. No audible knocking or hammering of mechanical parts shall occur in excess of that occurring in normal continuous operation as standard rating conditions in ARI 240, 320, 325, 340, and 380 as applicable.
e. Trip-out of operating, protective, and limit controls/devices shall not constitute a failure.
f. Average motor winding temperatures measured as described by the resistance method in IEEE 112 or 114, as applicable and shall not exceed 300°F (149°C) at any time during any reliability tests.
g. Discharge refrigerant line temperature within 6 inches of the compressor shall not exceed 250°F (121°C) during the reliability tests.
h. Upon completion of the reliability tests, the moisture content of the refrigerant shall be determined and the moisture content shall not exceed 30 ppm.

TABLE III. Reliability tests (high stress).

A. Air source heat pumps.
1. Heating and cooling test.
Voltage as specified in the contract.
Outdoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).
Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).
100-percent operation - 50-percent cooling and 50-percent heating.
Cycle reversal every hour.
48 hours total test duration.

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TABLE III. Reliability tests (high stress). (Continued)

2. High temperature test (cooling). Voltage 90 percent of specified. Outdoor air dry-bulb temperature 130°F (54.4°C) to 140°F (60.0°C). Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). 100-percent unit operation. 24 hours total test duration.
3. Low temperature test (heating). Voltage 110 percent of specified. Outdoor air dry-bulb temperature the higher of the lowest recorded temperature for the region or the unit cutoff temperature. Indoor dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). Operation: Off 24 hours, on 8 hours; off 24 hours, on 8 hours. 64 hours total test duration.
4. Fan motor failure or blocked air filter test (cooling). Voltage as specified. Outdoor air dry-bulb temperature 95°F (35.0°C) to 105°F (40.6°C). Indoor dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). 100-percent unit operation: 12 hours at 50 percent of normal air-flow across indoor coil, 12 hours at 0 percent of normal air-flow air-flow across indoor coil (indoor fan inoperative anf filter face blocked). 24 hours total test duration.
5. Fan motor failure or blocked air filter test (heating). Voltage as specified. Outdoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). Indoor dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). 100-percent unit operation: Inlet duct to be uniformly restricted at such a rate that the outlet air temperature increases a maximum of 1°F (0.6°C) per minute, until operation is interrupted by a temperature limit control or any other protective device. Once conditions have stabilized, increase the restriction until second temperature limit control functions, if provided. Procedure is repeated until the resulting temperatures and pressures are the maximum that occur with any amunt of restriction.
6. Low refrigerant charged test (cooling). Voltage as specified. Outdoor air dry-bulb temperature 95°F (35.0°C). Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). 50 percent of manufacturer's recommended refrigerant charge. 100-percent unit operation. 24 hours total test duration.

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TABLE III. Reliability tests (high stress). (Continued)

7. Low refrigerant charge test (heating).	
Conditions same as 5 above, except outdoor air temperature 20°F (-6.7°C) to 35°F (1.67°C).	
B. Water source heat pumps.	
1. Heating and cooling test.	
Voltage as specified in contract or purchase order.	
Entering water temperature 70°F (21.1°C) to 50°F (10.0°C).	
Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).	
100 percent operation: 50-percent cooling and 50-percent heating.	
Cycle reversal every hour.	
48 hours total test duration.	
2. High temperature test (cooling).	
Voltage 90 percent of specified.	
Entering water temperature 85°F (29.4°C) to 95°F (35.0°C).	
Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).	
100-percent unit operation.	
24 hours total test duration.	
3. Low temperature test (heating).	
Voltage 110 percent of specified.	
Outdoor air dry-bulb temperature the higher of the lowest recorded temperature for the region or the unit cutoff temperature.	
Entering water temperature 36°F (2.2°C) to 32°F (0°C).	
Indoor dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).	
Operation: Off 24 hours, On 8 hours, off 24 hours, on 8 hours.	
64 hours total test duration.	
4. Blocked air filter test (cooling).	
Voltage as specified.	
Entering water temperature 65°F (18.3°C) to 75°F (23.9°C).	
Indoor dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).	
100-percent unit operation: 12 hours at 50 percent of normal airflow across indoor coil, 12 hours with no airflow across indoor coil (indoor fan inoperative and filter face blocked for this test).	
24 hours total test duration.	
5. Fan motor failure or blocked air filter test (heating).	
Voltage as specified.	
Entering water temperature 70°F (21.1°C) to 80°F (26.7°C).	
Indoor dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C).	
100-percent unit operation: Inlet duct to be uniformly restricted at such a rate that the outlet air temperature increases a maximum of 1°F (0.6°C) per minute, until operation is interrupted by a temperature limit control or any other protective device.	
Once conditions have stabilized, increase the restriction until	

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TABLE III. Reliability tests (high stress). (Continued)

	second temperature limit control functions, if provided. Procedure is repeated until the resulting temperatures and pressures are the maximum that occur with any amount of restriction.
6.	Low refrigerant charge test (cooling). Voltage as specified. Entering water temperature 65°F (18.3°C) to 75°F (23.9°C). Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). 50 percent of manufacturer's recommended refrigerant charge. 100-percent unit operation. 24 hours total test duration.
7.	Low refrigerant charge test (heating). Conditions same as 5 above, except entering water temperature 45°F (7.2°C) to 55°F (1.28°C).

4.4.4.2 Life cycle reliability tests. Temperature tolerances for the reliability tests shall be as specified in performance requirements of the ARI 240 standard. Voltage tolerances shall be ±3 volts. Supplemental heaters shall not be operated during any reliability tests. Criteria for acceptance shall be those of 4.4.4.1 with the addition that trip out or similar operation of any manual-reset operating, protective, or limit control/device shall constitute a failure. Additional criteria for acceptance for the Life Cycle Reliability Test option shall be compressor mean time between failures of at least 38,990 hours of test time and system mean time between failures of at least 3,910 hours of test time, both for the 2,000-hour test cycle with 80-percent confidence, assuming a constant failure rate. Reliability computations are to take into account all failures without any exceptions A system failure is the occurrence of any condition which renders the heat pump system incapable of operating satisfactorily, within the specified performance limits, on a thermostat demand. A compressor failure constitutes a system failure requiring that the compressor be removed and replaced to restore satisfactory system operation. The minimum number of units to be tested to meet the above 80-percent confidence criteria may be calculated by satisfying equation 4.1 or may be read from table IV, which lists the minimum number of units to be tested for various numbers of compressor failures and the maximum number of unit failures consistent with each listed test sample size. The Life Cycle Reliability Tests are identified in table V.

$$\sum_{x=0}^{n} \frac{e^{-m} m^{x}}{x!} \leq 0.2 \qquad (4.1)$$

Where $m = \dfrac{\text{No. of units tested} * \text{No. of test hours per unit}}{\text{mean time between failures}}$

n = Total number of failures allowed.

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TABLE IV. Minimum number of test units and maximum number of compressor failures and system failures required for life cycle reliability testing.

Maximum Number of Compressor Failures	Minimum Number of Test Units	Maximum Number of Unit Failures
0	32	12
1	59	25
2	84	36
3	108	48
4	132	60

TABLE V. Reliability tests (life cycle).

1.	High temperature test (cooling). Outdoor air dry-bulb temperature 95°F (35.0°C) or higher. Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). Indoor relative humidity 50 percent. Operation: off 15 minutes/on 15 minutes. Voltage 110 percent of specified, 350 hours duration. Voltage 90 percent of specified, 350 hours duration.
2.	Defrost test (heating). Outdoor air dry-bulb temperature 38°F (3.3°C) to 32°F (0°C). Outdoor relative humidity 80 percent. Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). Operation: off 10 minutes/on 10 minutes. 800 hours duration.
3.	Low temperature test (heating). Outdoor air dry-bulb temperature the higher of the local design temperature region or the unit cutoff temperature. Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C). Operation: off 1 hour/on 1 hour. Voltage 110 percent of specified, 250 hours duration. Voltage 90 percent of specified, 250 hours duration.

4.4.4.3 Corrosion tests. Under extreme climatic conditions at the application site, the purchaser may find it appropriate to require the heat pump outdoor and/or indoor unit to pass the following tests.

4.4.4.3.1 Salt spray test. If a salt spray test is specified (see 6.2), portions of the heat pump that are normally exposed to the outdoor environment shall be tested in a corrosive environment as described in ASTM B 117 (salt spray test). Unless otherwise specified (6.2) order, the duration of this test shall be 240 hours.

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Salt spray test - ASTM B 117 (cooling operation 1 hour at conclusion).
Outdoor air dry-bulb temperature 97 ±2°F (36.1°C ±1.1°C) (atmospheric pressure in salt-spray chamber).
Indoor air dry-bulb temperature 70°F (21.1°C) to 80°F (26.7°C) (for 1 hour operation at conclusion of salt-spray test).
Compressor motor shall not be operated except for 1 hour at conclusion of salt-spray test.
Voltage as specified.
Outdoor unit fan 100-percent operation except as noted:
At the 24th and 96th hour of salt-spray exposure, the heat pump fan and salt spray are turned off and the chamber door is left open for a period of 8 hours.
At the end of the 8-hour period the test is resumed.
At the conclusion of the final hour of actual salt-spray exposure, the spray shall be discontinued, the chamber door opened, and the compressor turned on and operated in the cooling mode for 1 hour.
Total duration of actual salt-spray exposure shall be 240 hours (or 720 hours if specified in the contract or purchase order).

4.4.4.3.2 Indoor unit corrosion test. If the 100-percent relative humidity test is specified (see 6.2), the portions of the heat pump that are normally exposed to the indoor environment shall be tested as described in ASTM D 2247 (100 percent relative humidity test). The time duration of this test shall be 240 hours.

100-percent relative humidity test - ASTM D 2247
Indoor air temperature 85°F (29.4°C).
Indoor fan operation 100 percent (no heating or cooling coil operation).
Voltage as specified.
Interruption conditions at the 24th and 96th hour as specified in salt spray test of 4.4.4.3.1.
240 hours total 100-percent relative humidity exposure.
Flaking, perforation, or penetration of metal parts shall be considered a failure.

4.4.4.3.3 Criteria for acceptance. Data to be taken before and after corrosion tests shall be recorded and results photographed as given in section 3.3 of ASTM G 33. For painted and all coated surfaces, refer to ASTM D 1654 for testing procedure where marks shall be scribed on vertical, horizontal, external, and internal painted or coated surfaces. Rating Schedule No. 2 of ASTM D 1654 shall be reported. The rating number shall be not less than 6 and there shall be no rust creep greater than 1/8 inch (3.2 millimeters) normal to the scribe marks for acceptance. Blistering or flaking of paint or coating and rust creep at screw holes shall be reported and rated. Oxidation and corrosion of bare metal parts and at joints of dissimilar metals shall be reported and photographed. Flaking, perforation, or penetration of metal parts shall be considered a failure. Impairing damage to any mechanical or electrical components shall be considered a failure.

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4.5 Electromagnetic compatibility tests. When electromagnetic compatibility control is specified (see 3.27), the first article shall be tested in accordance with MIL-STD-462, to verify compliance with the pertinent provision of MIL-STD-461.

4.6 End item test. The heat pump shall be tested sufficiently to satisfy the manufacturer's quality assurance provisions and manufacturing and production test of UL 559. Failure to pass this test shall be cause for rejection.

4.7 Packaging inspection. The preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-R-3593.

5. PACKAGING

5.1 Preservation, packing, and marking. The preservation, packing, and marking shall be in accordance with the requirements of MIL-R-3593 with the levels of preservation and packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The heat pumps covered by this specification are intended for use as residential, commercial, or industrial units and are designed to perform the functions of either heating or cooling, circulating, filtering, cleaning, and dehumidifying an air stream with optional provisions for humidifying. The water-to-air units must be supplied with condenser water to carry away the heat of rejection when the units are on cooling or provide heat of absorption when the units are on heating. These heat pump systems utilize heat rejection equipment, such as closed evaporation coolers and cooling towers. This specification does not apply to room air conditioning type heat pumps as covered by the Association of Home Appliance Manufacturers.

6.1.1 Design conditions. Determine weather data, design, heating, and cooling loads and the design for air distribution system.

a. Weather data for outdoor design condition shall be in accordance with AFM 88-8, Chapter 6.
b. Design sensible heating and sensible/latent cooling loads shall be determined in accordance with procedures described in ASHRAE Handbook of Fundamentals or procedures given in or NESCA Manual J.
c. Except for closed-circuit heat pump systems, if water-to-air heat pumps are to be considered, an adequate proven year-round supply of water in gpm summer and winter water temperatures and water source shall be available. The quality (hardness, acidity, mineral content, and bacterial count) of the water and its temperature for both winter and summer seasons should be determined and stated in the ordering data in accordance with AWWA M12.

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d. Winter and summer design, indoor and outdoor conditions, and the seasonal heating hours for 5°F temperature increments below 65°F as given in the AFM 88-8, Chapter 6.
e. Required heating and cooling loads in Btu/hr, at specified winter and design conditions respectively.
f. Minimum external resistance of air distribution system when delivering rated capacity and air quantity, if different from the values listed in ARI 240.
g. For geographical regions of high corrosion potential, such as coastal regions with salt-spray exposure, specify salt-spray test (ASTM B 117) with 720 hours of exposure for portions of heat pump units exposed to the outdoor environment. For all other regions, specify salt-spray tests with 240 hours of exposure.
h. Air filters: Normally throwaway type filters should be specified; however, permanent type filters may be specified provided maintenance facilities are available for cleaning filters.
i. On type II systems, calculate length and specify (when required) interconnecting prefabricated factory charged tubing. Tubing should not exceed 100 feet in length.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, size, and capacity of heat pump required (see 1.2).
c. Issue of DODISS to cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When interconnecting refrigerant lines and length of line in feet are required (see 3.1.2 and 3.18).
e. When a first article is required for inspection and approval (see 3.4, 4.2.1, and 6.4).
f. When equipment, material, and articles are to be new and fabricated using materials produced from recovered materials (see 3.5).
g. When lifting attachments are required (see 3.7).
h. When overall dimensions are not to be determined by the manufacturer (see 3.10).
i. Whether couplings shall be brazed or sweat fitted or of the quick-connect type (see 3.10.1 and 3.18).
j. Static pressure and cfm required of fan (see 3.11).
k. When mechanical features to permit floor, ceiling, or other mounting provisions are required (see 3.12).
l. When outlet grilles are required (see 3.12.2).
m. Type of air filter required (see 3.14).
n. When time delay mechanism should be other than factory installed (see 3.15.4).
o. When refrigerant is to be other than R-22 (see page 3.16).
p. When the water-to-refrigerant heat exchanger, internal piping, and connection shall be other than a cupro-nickel alloy (see 3.22).
q. When provision for isolation and access valving is not required (see 3.22).

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r. For water-to-air heat pumps, the quantity (gpm) and the quality (hardness, acidity, mineral content, and bacterial count) of water and its temperature for both winter and summer available to the heat pump on a continuous basis (see 3.22 and 6.1.1(c)).
s. Electrical characteristics required (see 3.23.1).
t. When supplemental electric heating is required (see 3.23.2.1).
u. When capacity, kW, number of circuits, and stages specifically required (see 3.23.2.1).
v. When electrical duct heaters are required (see 3.23.2.1).
w. Nominal air flow requirement (see 3.23.2.2).
x. When temperature set-back for thermostat is required (see 3.23.3.3).
y. When outdoor thermostat setting other than 40°F (4.4°C) is required to raise balance point requirements (see 3.23.3.4).
z. The required heat pump heating and cooling capacity (sensible and latent) (see 3.24.1).
aa. When the minimum capacity of the supplemental heater shall be other than 80% of the design heating load (see 3.24.1)
ab. When one heat pump is required to undergo reliability tests (see 3.24.4, 4.2.2.4. and 4.4.4).
ac. Seasonal Performance Factor (SPF) for water source heat pumps and calculation methodology, if required (see 3.24.5.3).
ad. Required seasonal heating hours for 5°F temperature increments below 65°F to winter design temperature as given in AFM 88-8, Chapter 6, for location of installation of heat pumps (see 3.24.6).
ae. When minimum external duct resistance is other than specified in ARI 240 (see 3.24.7).
af. When fungus resistance is required (see 3.25).
ag. When treatment and painting is other than specified for indoor equipment (see 3.26).
ah. When electromagnetic compatibility is required (see 3.27).
ai. When marking of the Federal Stock Number is required (see 3.28i).
aj. When spare parts and maintenance tools are required (see 3.29).
ak. Sampling and inspection procedures shall be in accordance with level II of MIL-STD-105. The Acceptable Quality Levels shall be 1.5 percent defective for major defects and 4.0 percent defective of minor defects (see 4.2.2.2).
al. When one heat pump is required to undergo capacity and operational tests (see 4.2.2.3).
am. When test conditions for capacity tests shall be other than in ARI 240, 320, 325, 340, and 380 (see 4.4.2).
an. When one or both of the corrosion tests are to be specified (see 4.4.4, 4.4.4.3.1, 4.4.4.3.2)
ao. When duration of reliability test shall be other than 240 hours (see 4.4.4.3.1).
ap. Levels of preservation and packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List

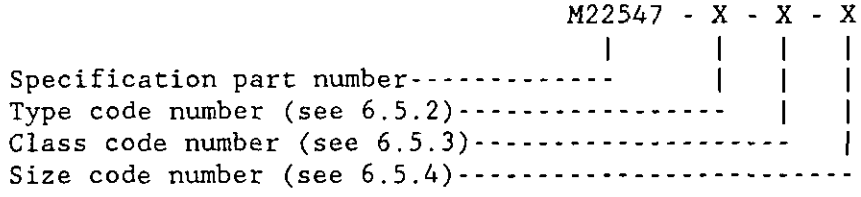
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(CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitive specification part or identifying number (PIN). The specification part or identifying number (PIN) is a definitive part number which corresponds to the type, class and size of Heat Pumps covered by this specification and defines the requirements of the options presented under this specification. The specification number, and the type, style, class, and arrangement code numbers are combined to form the definitive specification part number.

6.5.1 Cataloging data. For cataloging purposes, part numbers for the heat pumps are as follows:



6.5.2 Types. The types of the heat pumps (see 1.2) are identified by a one-digit number (see table VI).

TABLE VI. Code number to type.

Type I - 1
Type II - 2

6.5.3 Class. The classes of the heat pumps (see 1.2) are identified by a one-digit number (see table VII).

TABLE VII. Code number to class.

Class 1 - 1
Class 2 - 2

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6.5.4 Sizes. The sizes of the heat pumps (see 1.2) are identified by a one-digit number (see table VIII).

TABLE VIII. Code number size.

Size 1 - 1
Size 2 - 2

6.6 Subject term (keyword) listing.

Air conditioner
Heater

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Suppression of data. The document differs from the superseding specification in the following size classifications:

	MIL-H-22547B(YD)	MIL-H-22547C(YD)
Size 1	8,400 - 84,000 BTU/hr	8,400 - 65,000 BTU/hr
Size 2	85,000 - 300,000 BTU/hr	65,001 - 300,000 BTU/hr

Preparing Activity:

Navy - YD

(Project 4120-N316)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-22547C (YD)	Heat Pumps, Heating and Cooling (Unitary) (8,400 to 300,000 BTU)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 82 MAR

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